Crude and Product Terminals and Storage

Design Basis Accident — Military Action

DBA-OIL-FC3 — Facility Class Document

Issued Under: DBA-OIL Sector Framework

Version 0.1 — DRAFT

June 2026

Timothy E. Roxey

Eclectic Technologies

Prepared for: Petroleum Sector — Industry, Government, and Allied Partners

Parent document: DBA-OIL Sector Framework v1.4

Preface

The petroleum terminal is the least visible node in the petroleum supply chain and the most consequential one to lose. Refineries produce fuel. Pipelines move it. Terminals make the connection between the two — and between the pipeline system and the consumer. When a major product terminal cluster is destroyed, functioning refineries cannot deliver their output, and functioning pipelines have nowhere to deliver to. The supply disruption is immediate, regardless of the status of every other component in the chain. That is the terminal's strategic importance: it is the logistics bridge, and bridges are the first things armies destroy.

This document establishes the Design Basis Accident scenarios for deliberate military action against crude and product terminal and storage infrastructure. The facility class encompasses onshore crude storage terminals, petroleum product terminals, marine import and export terminals, and inland bulk liquid storage facilities. The primary geographic focus is the U.S. Gulf Coast product terminal complex — the origin point for the Colonial Pipeline and the distribution hub for approximately forty-five percent of U.S. refined product supply to the Eastern Seaboard — and the Northeast heating oil terminal network, whose seasonal demand profile creates a timing vulnerability that an adversary can exploit with precision unavailable in any other petroleum sector facility class.

The evidentiary anchors for this document are three. The Buncefield terminal fire (UK, 2005) is the governing reference event for the non-adversarial tank farm fire bounding case: a vapor cloud explosion and fire at a major product storage terminal destroyed twenty tanks, disrupted aviation fuel supply at Heathrow and Luton airports, and demonstrated how quickly a terminal fire propagates beyond the initiating tank. The Puerto Rico tank farm fire (Caribbean Petroleum Corporation, 2009) provides the boilover reference: eleven tanks were destroyed by a boilover event that escalated from a single initiating tank, overwhelming mutual-aid firefighting resources across the region. The Colonial Pipeline ransomware attack (2021) is the governing reference for the supply chain severance scenario: a six-day operational shutdown of the largest U.S. refined product pipeline produced regional fuel shortages across the Eastern Seaboard, demonstrating that the terminal-to-pipeline logistics interface is a consequence multiplier independent of physical terminal damage.

The DBA-MA extension builds on these references to define the bounding consequence when the initiating condition is not an accident but coordinated military action, with the adversary choosing the target, the timing, and the attack modality to maximize the duration and geographic scope of the supply disruption.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the crude and product terminals and storage facility class. Its purpose is to define the bounding threat event that terminal security planning, resilience investments, and recovery architecture must address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

A terminal security program designed against a single-vector physical threat — perimeter intrusion, vehicle-borne explosive — will fail when a sophisticated adversary combines cyber pre-positioning against the terminal's automation and safety systems with precision kinetic action against tank farm infrastructure and simultaneous attack against the marine berthing or pipeline connection infrastructure that links the terminal to the supply network. The DBA-MA approach defines the coordinated, multi-vector attack as the governing design event.

1.2 Scope and Applicability

This document addresses crude and product terminal and storage facilities: bulk liquid storage installations that receive, store, and distribute crude oil or refined petroleum products. The facility class encompasses onshore crude storage terminals that receive pipeline or marine crude and deliver to refineries; petroleum product terminals that receive refined products from refineries or pipelines and distribute to wholesale or retail customers; marine terminals with vessel berthing capability for tanker import or export operations; and inland bulk liquid storage facilities serving regional distribution functions.

Petroleum refining facilities are addressed in DBA-OIL-FC1. Offshore production platforms and associated export pipelines are addressed in DBA-OIL-FC2. The pipeline systems that connect terminals to refineries, offshore systems, and end markets are addressed in DBA-OIL-FC4. Where events at terminal facilities produce consequences that propagate into the pipeline system, this document addresses those consequences as they originate at the terminal without displacing the FC4 analysis. Where marine terminal events involve LNG, the DBA-GAS-UC-LNG use case governs the LNG-specific consequence analysis; this document addresses petroleum product marine terminals.

1.3 Relationship to Governing Documents

This document is issued under the DBA-OIL Sector Framework (v1.4), which is itself issued under the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the terminal-specific physical configuration, asset taxonomy, bounding events, and consequence chains that the sector framework establishes in general terms.

The inheritance map for this document is as follows. The threat actor taxonomy (Tier 1, 2, 3), the four characteristics of military action (intentionality, persistence, precision, denial), the three-tier consequence framework, and the adequacy standards are inherited unchanged from the sector framework. The bounding event categories are inherited with terminal-specific adaptation. The Dual Design Basis architecture and Bright Line / Command Broker requirements are inherited unchanged and applied to the terminal automation environment in Section 3.3.

1.4 The Terminal as the Logistics Bridge

A fundamental characteristic of the terminal and storage facility class distinguishes it from every other facility class in the DBA-OIL series: the terminal's strategic importance is defined not by what it produces but by what it connects. A refinery attack eliminates production. A pipeline attack severs transport. A terminal attack severs the connection between production and transport and between transport and consumption — simultaneously, regardless of the operational status of every other element in the chain.

This characteristic drives the entire DBA-MA analysis. An adversary targeting maximum supply disruption with minimum exposure does not need to attack a heavily guarded refinery or a remote offshore platform. A major product terminal cluster — accessible by road, visible from the air, operating under voluntary physical security standards with no mandatory regulatory equivalent to NERC CIP-014 — offers a more accessible and more consequence-efficient target. The terminal is the logistics bridge, and the adversary who destroys the bridge disconnects everything on both sides.

2. Terminal Profile and Asset Taxonomy

2.1 Terminal Types

Crude storage terminals. Bulk liquid storage facilities that receive crude oil from pipeline or marine sources and deliver it to refineries. Crude terminals typically hold several million barrels of storage capacity in large floating-roof tanks ranging from 500,000 to 1,000,000 barrels each. The Cushing, Oklahoma, terminal complex — the delivery point for the WTI crude oil benchmark — holds approximately 80 million barrels of storage capacity and is the most consequential crude storage node in the U.S. pipeline system. Gulf Coast crude terminals receive domestic and imported crude for delivery to the Gulf Coast refinery cluster; their incapacitation affects refinery feedstock supply independent of offshore or onshore production status.

Petroleum product terminals. Bulk liquid storage and distribution facilities that receive refined products from pipelines or refineries and distribute to wholesale customers — truck loading, rail, and barge. Product terminals hold gasoline, diesel, jet fuel, and heating oil in separate dedicated tanks. Terminal tank farms typically range from 500,000 to 5,000,000 barrels of total product storage capacity. Product terminals at the origin points of major pipeline systems — the Gulf

Coast terminals that feed Colonial Pipeline — are the highest-consequence nodes in the refined product distribution chain.

Marine terminals. Terminals with deepwater berths capable of receiving crude tankers or product tankers for import or export operations. Marine terminals combine the asset profile of the onshore tank farm with the maritime infrastructure of jetties, loading arms, and mooring systems. The LOOP (Louisiana Offshore Oil Port) is the only U.S. deepwater offshore crude import terminal; Gulf Coast marine product terminals handle both import and export operations for refined products. Northeast marine terminals receive product tanker imports that supplement Colonial Pipeline supply, particularly for heating oil.

Inland bulk liquid storage. Regional distribution terminals away from major pipeline corridors are served by barge, rail, or truck. Inland terminals buffer the last-mile distribution system; their loss affects local markets rather than regional or national supply chains. They are addressed in this document as the downstream consequence receiver for Scenario C rather than as primary attack targets.

2.2 Tank Types and Physical Configuration

Floating-roof tanks. The dominant tank type for volatile petroleum products — gasoline, crude oil, and intermediate products. A floating roof rides on the liquid surface, eliminating the vapor space above the product and reducing evaporative emissions and fire risk. Floating-roof tanks are the primary fire risk at product terminals: a rim seal fire — ignition of the vapor at the seal between the floating roof and the tank shell — is the most common tank fire initiating event. Rim seal fires are manageable with prompt response; the bounding case is a seal fire that escalates when the roof sinks or tilts, creating a full-surface fire in a large-diameter tank where foam application is operationally challenging.

Fixed-roof tanks. Used for less volatile products — diesel, heating oil, jet fuel — and for crude storage where vapor pressure permits. Fixed-roof tanks maintain a vapor space above the liquid surface; the vapor-air mixture in that space is potentially flammable. A fixed-roof tank fire typically results from the ignition of vapors vented from the tank during filling operations or from a leak. The boilover scenario applies primarily to fixed-roof crude tanks: water accumulated at the bottom of the tank is superheated by the burning surface layer and erupts violently, projecting burning crude over the tank walls and beyond the dike — the mechanism responsible for the most severe escalation events in tank farm fire history.

Pressurized storage. Spheres and bullets for LPG, propane, and other pressurized products. Pressurized storage vessels present a BLEVE risk — Boiling Liquid Expanding Vapor Explosion — if exposed to fire conditions. BLEVE events are the highest-consequence single-vessel failure mode at a petroleum terminal; their consequence envelope extends well beyond the terminal boundary. Pressurized storage is typically a smaller component of product terminal capacity but represents a disproportionate safety consequence risk.

2.3 Major Terminal Clusters

Gulf Coast — Colonial Pipeline origin complex. The product terminal cluster at the Colonial Pipeline origin points in Houston, Pasadena, and Port Arthur, Texas, represents the most consequential single product terminal concentration in the United States. Colonial Pipeline carries approximately 100 million gallons per day of gasoline, diesel, and jet fuel from these terminals to markets along the Eastern Seaboard, serving approximately forty-five percent of the Eastern U.S. refined product demand. A successful attack on the Gulf Coast terminal complex that forces a Colonial Pipeline shutdown replicates the 2021 ransomware consequence — Eastern Seaboard fuel shortages within days — with physical damage that extends the outage from days to months.

Gulf Coast — marine crude terminals. The crude import marine terminal network along the Texas and Louisiana coast, including the LOOP, receives foreign crude that supplements domestic production for the Gulf Coast refinery cluster. LOOP is the only terminal capable of receiving Very Large Crude Carriers (VLCCs) at an offshore buoy, with a pipeline connection to onshore storage. LOOP handles approximately 10 to 15 percent of U.S. crude oil imports; its loss forces crude importers to use smaller tankers with longer transit times and higher costs, reducing import throughput.

Northeast — heating oil terminal network. The New England and Mid-Atlantic coastal product terminal network receives heating oil, diesel, and gasoline by marine tanker from Gulf Coast refineries and serves the residential and commercial heating market that is uniquely dependent on distillate fuels. Massachusetts, Maine, New Hampshire, Vermont, Connecticut, and Rhode Island have limited pipeline connectivity to the Gulf Coast supply system; their primary supply path is marine tanker to coastal terminal to retail delivery truck. The Northeast heating oil terminal network is the highest-consequence seasonal target in the DBA-OIL series: an attack timed for the onset of heating season eliminates the buffer inventory precisely when demand is accelerating.

West Coast — Los Angeles Basin terminals. Product terminals in the Los Angeles Basin serve the West Coast market, which has minimal pipeline interconnection to other refining regions. West Coast product supply is almost entirely dependent on local refinery production; a terminal attack that disrupts local refinery-to-terminal delivery forces immediate retail supply stress without the alternative supply routing that the East Coast can achieve through marine imports.

2.4 Regulatory Authority

EPA SPCC. The Spill Prevention, Control, and Countermeasure rule (40 CFR Part 112) requires facilities that store oil above threshold quantities to prepare and implement spill prevention plans, maintain secondary containment, and document inspection and testing programs. SPCC is the primary environmental protection regulatory framework for petroleum terminal operations and establishes the baseline adequacy standard for spill response capability. SPCC does not address security; it addresses accidental release prevention.

USCG. The United States Coast Guard (33 CFR) has jurisdiction over marine terminal security under the Maritime Transportation Security Act (MTSA). Marine terminals that handle petroleum products above threshold quantities are required to develop and implement Facility Security Plans (FSPs) reviewed and approved by the USCG. The MTSA framework provides the closest analog to a mandatory security standard for this facility class. Still, it applies only to the marine interface — the waterside security of berthing and transfer operations — not to the onshore tank farm.

PHMSA. The Pipeline and Hazardous Materials Safety Administration (49 CFR Part 195) regulates the pipeline connections between terminals and the trunk pipeline system. The terminal's internal piping and pumping systems are generally subject to EPA and OSHA process safety standards rather than PHMSA pipeline regulations. The boundary between PHMSA pipeline jurisdiction and terminal operator jurisdiction is a regulatory gap that affects incident response coordination.

TSA. TSA Pipeline Security Directives, issued following the Colonial Pipeline ransomware attack, establish mandatory cybersecurity requirements for pipeline owners and operators. These directives apply to the pipeline systems connected to terminals, not to the terminals themselves. Terminal operators whose facilities are not pipeline-owner entities may not be subject to TSA cybersecurity directives, creating a gap between the pipeline security framework and the terminal security framework at the precise point of their interconnection.

3. Design Basis: Crude and Product Terminals and Storage Facility Class

3.1 Non-Adversarial Bounding Events

The non-adversarial Design Basis for the terminal and storage facility class inherits the process safety event, infrastructure failure event, natural hazard event, and operational event categories from the DBA-OIL Sector Framework (Sections 4.3 through 4.5). This document develops the facility-class-specific consequence chains for three governing non-adversarial bounding events.

Tank farm fire and boilover. The governing process safety bounding event is a major tank farm fire with boilover escalation. The Buncefield terminal fire (UK, 2005) is the primary reference event for the vapor cloud explosion and fire progression: a floating-roof tank overflow during filling produced a large vapor cloud that ignited, causing an explosion that ignited multiple tanks and destroyed approximately twenty storage tanks holding gasoline and aviation fuel. Aviation fuel supply at Heathrow and Luton airports was disrupted for several days, and the surrounding area sustained blast damage across a significant radius. The Puerto Rico boilover (Caribbean Petroleum Corporation, 2009) is the primary boilover reference: a tank fire escalated to boilover, projecting burning crude beyond the dike, igniting adjacent tanks, and ultimately destroying eleven tanks in a fire that burned for several days and overwhelmed regional firefighting resources. The compound consequence — physical destruction of storage capacity, product loss, environmental contamination from firefighting runoff, and supply chain disruption — defines the non-adversarial bounding envelope.

Large product spill — marine terminal. Loss of containment at a marine terminal during tanker transfer operations, resulting in a large-volume product release to navigable waters. The bounding case involves a transfer line failure or loading arm disconnection during high-flow transfer operations, producing a release that exceeds the terminal's containment capacity and reaches the waterway. The regulatory response — USCG Federal On-Scene Coordinator activation, EPA response, PHMSA notification — is an embedded component of the consequence envelope that extends the operational disruption significantly beyond the physical repair timeline.

Control system failure — terminal automation. Loss of the terminal automation system (TAS) or SCADA system governing tank level management, pump operations, and product transfer, forcing manual operations across the terminal. A TAS failure during high-throughput operations can result in tank overfills, incorrect product routing, or pump cavitation damage. Manual operation of a large product terminal is operationally feasible but significantly reduces throughput — pipeline delivery scheduling cannot be maintained at normal rates during manual TAS operation, creating a supply chain ripple effect that extends beyond the terminal's own storage capacity.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the terminal and storage facility class is a coordinated, multi-vector attack against a major product terminal cluster or marine terminal, executed during a high-demand or high-inventory period, designed to simultaneously destroy storage capacity, sever the pipeline and marine supply interface, and extend the recovery timeline by targeting the longest-lead-time recovery components.

The bounding parameters, inherited from the sector framework threat taxonomy:

Adversary tier: Tier 1 state actor with cyber pre-positioning capability and precision kinetic or UAS delivery. The terminal target set is also accessible to a Tier 2 near-state proxy — the physical security standards are voluntary, and the facilities are not hardened against a determined, equipped adversary.

Modality: Cyber-kinetic combined arms. Terminal automation system compromise suppresses tank level alarms, inhibits emergency shutdown of transfer operations, and manipulates product routing to create overfill conditions in targeted tanks. Kinetic action — precision munitions, UAS-delivered charges, or vehicle-borne explosive — against tank farm infrastructure executes within the alarm-suppression window, simultaneously targeting the marine berthing infrastructure and the pipeline meter station to sever both supply interfaces.

Target selection: The pipeline origin terminal cluster whose loss produces the maximum downstream supply disruption. For the Eastern Seaboard supply chain, that is the Gulf Coast Colonial Pipeline origin complex. For the Northeast heating market, that is the Boston Harbor / Providence / New Haven coastal terminal cluster. Target selection is informed by the adversary's knowledge of inventory levels, pipeline schedules, and the seasonal demand curve — information available through open-source shipping and pipeline flow data.

Timing: The onset of peak demand season — late November for the Northeast heating market, Memorial Day weekend for the Gulf Coast gasoline market — when terminal inventories are high (maximizing the physical destruction consequence) and demand is about to accelerate (minimizing the time available for the supply chain to adapt before retail shortages appear).

Simultaneity: Coordinated attacks on two or more terminal facilities within the same supply chain corridor within a window that prevents the pipeline operator from re-routing supply before the retail market registers the disruption.

3.3 Dual Design Basis Architecture

Consistent with the DBA-OIL Sector Framework (§5.4) and DBA-EN-SF (§4.2), a petroleum terminal facility may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.1 governs the boundary between the two design bases.

Three terminal-specific AI governance interaction points are identified for the FC3 Design Basis:

Terminal automation system and tank gauging. TAS systems that continuously manage tank levels, product routing, and transfer pump operations are ML systems operating below the Bright Line when performing deterministic level management and pump control functions. They are formally testable within their operating envelopes and may operate autonomously. However, a TAS that is adversarially manipulated — suppressing high-level alarms, masking product routing errors, or inhibiting pump shutdowns — is functionally equivalent to adversary control of the terminal's product inventory. The AI Governance Design Basis must establish that TAS systems are architecturally isolated from the corporate IT network, that tank level data from independent gauging systems is compared against TAS readings before any high-volume transfer is authorized, and that anomalous level trajectories trigger local alarms independent of the TAS.

Pipeline scheduling and nomination systems. Pipeline operators use scheduling and nomination systems to manage product delivery schedules across the terminal-to-pipeline interface. Gen AI advisory tools for pipeline scheduling optimization — tools that recommend delivery windows, batch sequences, and product allocations — are increasingly deployed at major terminal complexes. These tools operate above the Bright Line: they are non-deterministic and not formally testable. A human Command Broker must be in the control loop for any pipeline delivery schedule change recommended by a Gen AI scheduling tool. A Gen AI tool whose recommendations are automatically translated into pipeline nominations without human review violates the Command Broker architecture, and in an adversarial context, a compromised Gen AI scheduling tool becomes a mechanism for manipulating pipeline flow at the terminal interface.

Fire and gas detection system ML. Fixed fire and gas detection systems at terminal facilities incorporate ML-based signal processing to reduce false alarm rates and improve detection sensitivity. These systems are ML operating below the Bright Line when performing signal classification functions within their trained envelopes. The adversarial concern is not the autonomous operation of the detection system but the manipulation of the ML model — through adversarial inputs to training data or through direct model parameter manipulation — to suppress detection of an actual release or fire. The Design Basis requirement is that fire and gas detection system ML models are validated against a defined set of adversarial input scenarios, that model update processes are subject to the same change management controls as the underlying safety system hardware, and that manual detection capability — human rounds and independent fixed detection — is maintained as a backup to the ML-augmented system.

4. Design Basis Accident Scenarios

4.1 Scenario A: Coordinated Attack on the Gulf Coast Product Terminal Complex

4.1.1 Initiating Event

Coordinated cyber-kinetic attack against the Colonial Pipeline origin terminal cluster in the Houston Ship Channel / Pasadena / Port Arthur corridor. The cyber component, having achieved TAS access through a multi-stage intrusion campaign exploiting the terminal's IT-OT network boundary over the preceding months, initiates simultaneous manipulation of tank level management across multiple gasoline and diesel storage tanks — suppressing high-level alarms and inhibiting transfer pump shutdowns at three to four tanks approaching high-level. The TAS manipulation is timed to occur during a high-throughput pipeline delivery window when multiple tanks are receiving simultaneous pipeline injections.

The kinetic component executes within the alarm-suppression window: UAS-delivered charges against two floating-roof gasoline tanks — initiating rim seal fires — and a vehicle-borne explosive against the Colonial Pipeline meter station and pump isolation valves at the terminal boundary. The tank fires and the meter station destruction occur within minutes of each other.

4.1.2 Progression Sequence

T+0: Rim seal fires initiated at two floating-roof gasoline tanks. Meter station destroyed, severing Colonial Pipeline's physical connection to the terminal complex. TAS alarm suppression prevents immediate detection of the overfill conditions developing in three additional tanks. Terminal fire brigade responds to the rim seal fires.

T+0 to T+30 minutes: Rim seal fires escalating as TAS manipulation continues to pump product into the affected tanks. One floating roof begins to tilt under uneven product loading — transitioning from a manageable rim seal fire to a full-surface fire. Adjacent tanks are at risk from radiant heat. Mutual aid firefighting resources are requested. Colonial Pipeline shutdown initiated from the control center in response to meter station destruction — the entire Colonial system is offline.

T+30 minutes to T+4 hours: Full-surface fire on the tilted floating-roof tank producing intense radiant heat that threatens the adjacent tank farm. Mutual aid foam resources are exhausted on the primary tank; secondary tanks are receiving radiant heat exposure. TAS forensic investigation initiated — extent of compromise unknown. Colonial Pipeline shutdown confirmed; DOE ESF-12 notification initiated. Product routing to the alternative terminal or the marine import assessment is underway.

T+4 to T+72 hours: Major fires knocked down after mutual aid foam application from multiple regional fire agencies. Physical damage assessment: two tanks destroyed, two tanks with structural damage requiring inspection before return to service, meter station destroyed. Product loss of approximately 2 to 4 million barrels from destroyed and contaminated tanks. Environmental contamination from firefighting runoff exceeding secondary containment capacity — EPA Federal On-Scene Coordinator activated. Colonial Pipeline remains offline pending meter station replacement and security assessment.

T+72 hours onward: Colonial Pipeline meter station procurement initiated — lead time assessment indicates 8 to 16 weeks for a custom meter station with the required throughput capacity. TAS forensic investigation and reconstitution are underway — a minimum of 4 to 6 weeks before the TAS can be trusted for automated terminal operations. Eastern Seaboard fuel shortages are developing, beginning in the Southeast within 72 hours and reaching the Mid-Atlantic and New England markets within 5 to 7 days.

4.1.3 Consequence Envelope

Immediate supply impact. Colonial Pipeline carries approximately 100 million gallons per day of gasoline, diesel, and jet fuel to Eastern Seaboard markets. A shutdown of more than five days produces measurable retail fuel shortages in Southeast markets; a shutdown of more than ten days produces shortages in Mid-Atlantic markets; a shutdown of more than two weeks reaches New England. The 2021 ransomware event demonstrated this progression over six days from a voluntary shutdown — the DBA-MA scenario produces a shutdown whose duration is governed by the meter station replacement timeline, not by a recovery decision.

Recovery timeline. The governing constraint is the meter station replacement. A custom pipeline meter station — designed for the specific throughput, pressure rating, and product mix of the Colonial Pipeline origin point — has a procurement and installation timeline of 8 to 16 weeks under normal supply chain conditions. TAS reconstitution runs in parallel but must be complete before full automated operations can resume. The compound effect produces an Eastern Seaboard refined product supply disruption of 2 to 4 months' duration — qualitatively different from the 6-day Colonial Pipeline event and beyond the SPR's product release capability, which is limited to crude oil, not refined products.

Military logistics impact. Military aviation fuel supply at bases along the Eastern Seaboard is drawn from the same Colonial Pipeline product stream. A 2 to 4 month Colonial Pipeline shutdown requires DOD to activate emergency fuel logistics — alternative supply routing, strategic petroleum reserve crude release to sustain Gulf Coast refinery throughput, and priority allocation of available product to military consumers. This activation competes with civilian emergency allocation decisions and creates interagency coordination demands that compound the operational consequence.

4.2 Scenario B: Marine Terminal Attack — Tanker and Terminal Combined

4.2.1 Initiating Event

Coordinated attack against a major Gulf Coast marine crude terminal and an inbound VLCC at anchor awaiting berth. The attack uses the maritime operational template established by the Houthi campaign: a combination of USV (uncrewed surface vessel) attack against the tanker hull and a UAS strike against the terminal's marine loading arm and jetty infrastructure. The tanker attack is timed to occur when the vessel is at anchor with a full cargo of approximately two million barrels of crude — maximizing the spill consequence — while the terminal attack simultaneously destroys the berthing and offloading infrastructure that would receive the cargo.

4.2.2 Compound Consequence: Vessel and Terminal

The compound attack produces two simultaneous consequence chains that interact and amplify each other. The tanker hull breach initiates a major crude oil spill in the approach channel or anchorage area — potentially two million barrels if the cargo tanks are breached, far exceeding any USCG or industry pre-positioned response capability. The terminal infrastructure destruction eliminates the facility's ability to receive any vessel — including emergency response vessels — at the berth, and destroys the loading arm and jetty infrastructure whose replacement has a lead time of 6 to 18 months.

The crude oil spill in the approach channel creates a secondary consequence: navigation closure. A major spill in a ship channel requires USCG closure of the waterway to vessel traffic until the spill is sufficiently contained for safe navigation. For Gulf Coast terminals situated in the Houston Ship Channel, a navigation closure affects not only the attacked terminal but every marine terminal along the channel — including terminals unrelated to petroleum that serve other sectors. The channel closure duration following a two-million-barrel spill in confined waters is measured in weeks to months, not days.

4.2.3 Consequence Envelope: Marine Terminal Attack

Spill consequence. A two-million-barrel crude oil spill in a confined ship channel is the largest non-well-blowout spill scenario in the DBA-OIL series. The USCG's National Response System has a maximum credible spill planning basis of approximately 200,000 barrels for the Gulf Coast; the DBA-MA scenario exceeds this by an order of magnitude. Environmental consequences — shoreline contamination, fishery closure, and long-term remediation — extend across the affected waterway system for years following the initiating event. The economic and political consequences of a visible, large-scale marine spill compound the supply disruption consequences.

Terminal recovery timeline. Marine loading arm and jetty reconstruction has a lead time of 6 to 18 months, depending on the extent of structural damage to the jetty foundation. The terminal cannot receive any vessel until the berthing infrastructure is restored; marine crude imports to the affected terminal cease entirely for the duration of reconstruction. The crude import capacity diverted from the attacked terminal must be absorbed by alternative terminals, creating congestion and throughput constraints across the Gulf Coast marine crude import system.

Navigation closure cascade. The ship channel navigation closure affects all marine traffic in the affected waterway, creating a multi-sector consequence that extends beyond the petroleum supply chain. Petrochemical plant feedstock deliveries, export grain movements, and non-petroleum import cargo are all disrupted by the channel closure. The cross-sector consequence is a Tier 3 event in the DBA-OIL series consequence framework.

4.3 Scenario C: Northeast Heating Oil Terminal Attack — Seasonal Timing as a Force Multiplier

4.3.1 Initiating Event

Coordinated attacks against three major heating oil terminals in the Boston Harbor / Providence / New Haven coastal cluster, executed in the last week of November — the onset of the Northeast heating season, when terminal inventories are at or near maximum capacity following the summer fill season and residential heating demand is beginning its seasonal acceleration. The attack uses the same cyber-kinetic combined-arms modality as Scenario A: TAS compromise to suppress alarms and inhibit emergency shutdown, followed by kinetic action — UAS or standoff precision munitions — against tank farm infrastructure and marine berthing at each terminal.

4.3.2 Seasonal Timing as the Governing Force Multiplier

The Northeast heating oil market has structural supply chain characteristics that make the timing of an attack as consequential as its physical scale. The Northeast receives the majority of its heating oil supply by marine tanker from Gulf Coast refineries; the Colonial Pipeline provides supplemental supply to the Mid-Atlantic but does not reach New England. The regional storage terminals are the primary buffer between the marine supply chain — tankers that require 7 to 14 days in transit from Gulf Coast refineries — and the residential delivery system that requires product on a 24-to-48-hour replenishment cycle during peak demand.

An attack at the onset of heating season exploits three simultaneous vulnerabilities: inventories are high, maximizing the physical destruction consequence in product volume destroyed; demand is accelerating, reducing the time before retail shortages appear; and the marine supply replenishment cycle — 7 to 14 days from order to delivery — cannot be compressed even if alternative terminals remain operational. The adversary who understands these dynamics does not need to destroy all Northeast heating oil terminals. Destroying sufficient terminal capacity to reduce regional storage below the demand-season buffer requirement produces heating oil shortages within two to three weeks, regardless of the physical status of the remaining terminal infrastructure.

4.3.3 Consequence Envelope: Northeast Heating Oil

Residential heating impact. Approximately 5.5 million households in the Northeast depend on heating oil as their primary space heating fuel. At peak heating demand — a sustained cold spell in January or February — the region consumes approximately 800,000 to 1,000,000 barrels of heating oil per day. A terminal capacity reduction that prevents the regional storage system from sustaining this demand rate produces heating oil shortages within 2 to 3 weeks of the attack. Residential heating shortages in winter — particularly for elderly populations in rural areas dependent on heating oil — are a direct life-safety consequence.

Emergency response constraint. The Northeast heating oil supply chain has no rapid alternative. There is no pipeline delivering heating oil to New England from outside the region. Alternative marine supply routing — from European refineries or alternative U.S. refinery regions — requires 2 to 4 weeks of transit and pre-arrangement. The Strategic Petroleum Reserve contains crude oil, not heating oil; its drawdown assists refineries but does not directly supply the residential heating market. The gap between the attack consequence and the available emergency response mechanisms is wider for this scenario than for any other in the DBA-OIL series.

Political consequence. A residential heating shortage in a Northeast winter is among the highest-visibility domestic energy security failures in the U.S. political environment. The political pressure generated by heating oil shortages — particularly when they follow a demonstrably successful adversary attack — creates pressure for reactive regulatory and policy responses that may be poorly designed. The adversary who understands U.S. domestic political dynamics may value the political consequence of this scenario as highly as its supply disruption consequence.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in U.S. petroleum terminal infrastructure or specific terminal facility layouts. Anomalous TAS network activity: unusual access to tank gauging data, anomalous pipeline scheduling system queries, unauthorized access attempts to the IT-OT boundary. Physical surveillance of terminal perimeters or marine approaches. Unusual aerial or maritime activity in terminal waterway corridors or airspace. Procurement inquiries for terminal engineering data, pipeline meter station specifications, or storage tank design documentation—open-source reporting of elevated adversary maritime activity in Gulf Coast waters or Northeast coastal approaches.

5.1.2 Imminent Threat Indicators

Confirmed unauthorized UAS in terminal airspace or waterway approaches. Simultaneous TAS anomalies at multiple terminals within a regional supply chain corridor—confirmed unauthorized vessel in the terminal marine security zone. Unexplained tank level anomalies at multiple tanks are inconsistent with scheduled delivery operations. Confirmed kinetic event at any petroleum terminal facility. Simultaneous security incidents at two or more terminals within a pipeline corridor within 48 hours.

5.2 Decision Authority Framework

Facility level: Terminal operator authority to initiate emergency shutdown of transfer operations, isolate tanks, and activate fire suppression systems. USCG notification for marine incidents. EPA notification for spills reaching or threatening navigable waters. TSA and CISA notification for confirmed or suspected cyber incidents affecting terminal automation systems. Local emergency management notification for events with potential off-site consequences.

Federal coordination: USCG Federal On-Scene Coordinator for marine spill events. EPA Federal On-Scene Coordinator for onshore spills with environmental consequences. DOE ESF-12 for supply disruption response. CISA for cyber incident coordination. FBI and DOD for events with confirmed or strongly suspected state-actor attribution.

Pipeline operator coordination: Pipeline operators whose systems connect to attacked terminals must be notified immediately; the pipeline shutdown decision — which has the largest downstream consequence — may be made by the pipeline operator independent of the terminal operator's emergency response actions. Formal notification protocols between terminal operators and connected pipeline operators are a current gap in the emergency response framework.

5.3 Federal Coordination Thresholds

Confirmed kinetic strike or deliberate fire initiation at any major petroleum product terminal. Confirmed TAS compromise with suspected nation-state attribution at any petroleum terminal. Simultaneous security incidents at two or more terminals within a single pipeline supply corridor within 48 hours. Marine spill affecting or threatening to affect a navigable waterway above a defined volume threshold. Supply disruption affecting Colonial Pipeline or an equivalent major product pipeline system. Heating oil terminal capacity loss is affecting the Northeast winter supply buffer below the defined threshold during the heating season.

6. Recovery Architecture

6.1 Terminal Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Emergency shutdown and fire suppression. Environmental spill containment and USCG/EPA Federal On-Scene Coordinator activation for marine events. TAS forensic investigation initiation. Pipeline operator notification and coordinated shutdown decision. Alternative product supply routing assessment — marine import surge, alternative terminal activation, emergency product import authorization. DOE ESF-12 activation for supply disruption affecting national markets. SPE drawdown authorization for crude supply emergency to maintain refinery throughput.

Phase 2 — Short-term (72 hours to 8 weeks). Physical damage assessment. TAS forensic investigation complete; systems reconstituted before any automated terminal operations resume. Tank structural assessment — undamaged tanks returned to service as TAS integrity is confirmed. Meter station and pipeline connection repair or replacement procurement initiated. Marine berthing temporary repair assessment — whether a temporary berth can be established to receive product tankers while permanent reconstruction proceeds. Alternative supply arrangements formalized — marine import volume, alternative pipeline routing, and emergency product release from DOD reserves.

Phase 3 — Medium-term (8 weeks to recovery). Tank farm reconstruction where required. Marine infrastructure permanent reconstruction. Pipeline meter station replacement. Sequential return to service following regulatory inspection. Supply chain monitoring for adversary interdiction of replacement terminal components — tank shells, loading arms, and meter station components have lead times that can be extended by adversary supply chain interdiction.

6.2 Critical Recovery Constraints

Pipeline meter station replacement. A custom pipeline meter station for a major pipeline origin point is engineered to the specific pipeline's pressure rating, throughput capacity, and product mix. Replacement lead times of 8 to 16 weeks under normal conditions can extend further if the adversary has interdicted the supply chain for precision metering equipment. No strategic reserve of pipeline meter stations exists in the U.S. petroleum infrastructure.

Marine loading arm and jetty reconstruction. Marine loading arm systems — the articulated arm structures that connect the terminal piping to the vessel manifold — are custom-engineered to the terminal's berth geometry and product transfer rate. Replacement lead times of 6 to 18 months reflect both the manufacturing lead time and the structural engineering required for jetty reconstruction. Temporary berth solutions — mooring dolphins, floating hose transfer systems — can restore partial marine import capability within weeks, but at significantly reduced throughput.

Foam and firefighting resource depletion. A major tank farm fire at a large product terminal can consume the regional mutual-aid foam supply — particularly Class B firefighting foam for large-diameter petroleum tank fires — within hours. Foam resupply from national stockpiles requires days to a week. A multi-terminal attack timed to exhaust the regional foam supply before the second terminal fire is initiated exploits this constraint directly. No national strategic reserve of large-tank firefighting foam exists in a form pre-positioned for rapid deployment.

TAS reconstitution. Terminal automation system forensic investigation and reconstitution must be completed before any automated tank management can resume. A minimum of 4 to 6 weeks for a major TAS forensic investigation and reconstitution — similar to the DCS reconstitution timeline in FC1 — runs in parallel with physical repair but gates the restart of automated high-throughput operations regardless of physical repair status.

6.3 Northeast Heating Oil Emergency Supply

The Scenario C consequence envelope requires a dedicated recovery architecture element that has no equivalent in the other FC3 scenarios: the emergency marine supply surge for the Northeast heating market. The Northeast has no pipeline alternative to marine supply; the recovery mechanism is therefore entirely dependent on the availability of product tankers willing to load heating oil at Gulf Coast or other refineries and transit to New England ports, and on the operational status of the receiving terminals that were not attacked.

The current regulatory framework for emergency heating oil supply — the Northeast Home Heating Oil Reserve (NHOR), which holds approximately one million barrels of heating oil — provides approximately one day of peak-demand supply for the region. The NHOR is not sized for the Scenario C consequence envelope. Emergency import authorization — waiving the Jones Act for foreign-flagged tankers to deliver heating oil between U.S. ports — provides an additional lever that has been used in previous supply emergencies but requires presidential authorization and takes days to implement after the emergency is declared.

7. Gaps and Recommendations

7.1 Regulatory Gap: Absence of Mandatory Physical Security Standards for Onshore Terminals

Onshore petroleum product terminals — including the Colonial Pipeline origin terminal complex that supplies forty-five percent of Eastern Seaboard refined product — have no mandatory physical security standard. CFATS covers some chemicals of interest; the MTSA covers marine terminal waterside security; SPCC covers spill prevention. None of these frameworks requires a physical security assessment against the Design Basis Threat defined in this document. A terminal whose destruction would produce a multi-month Colonial Pipeline shutdown has no regulatory requirement to conduct a security vulnerability assessment, implement perimeter intrusion detection, or assess its vulnerability to UAS attack. This gap is the direct analog of the regulatory gap identified in DBA-OIL-FC1 §7.1. It reflects a sector-wide failure to extend mandatory security standards beyond the pipeline to the terminal infrastructure on which the pipeline depends.

7.2 Terminal Automation System Cybersecurity Gap

TSA Pipeline Security Directives, issued following the Colonial Pipeline ransomware attack, establish mandatory cybersecurity requirements for pipeline owners and operators. These directives do not cover terminal operators whose facilities are not pipeline-owner entities. The TAS systems governing tank level management at major product terminals — systems whose compromise is the cyber component of the Scenario A bounding event — have no mandatory cybersecurity framework. TSA should evaluate extending the pipeline cybersecurity directive framework to cover terminal automation systems at terminals connected to or serving major pipeline systems, beginning with the Colonial Pipeline origin terminal complex.

7.3 Northeast Heating Oil Reserve Adequacy

The Northeast Home Heating Oil Reserve holds approximately one million barrels — one day of peak regional demand. The Scenario C consequence envelope produces a multi-week heating oil supply disruption. The reserve is not sized for the design-basis military-action scenario. DOE, in coordination with the Northeast state energy offices and the heating oil distribution industry, should assess the reserve volume required to bridge the gap between a Scenario C attack and the marine supply surge that represents the primary recovery mechanism. The assessment should quantify the emergency marine supply surge timeline and establish the reserve volume required to sustain residential heating without interruption during that interval.

7.4 Firefighting Foam Strategic Reserve

No national strategic reserve of Class B firefighting foam for large petroleum tank fires exists in a form pre-positioned for rapid regional deployment. The mutual-aid foam supply that a major product terminal cluster can access is sized for a single large-tank event; the Scenario A and Scenario C bounding events involve simultaneous multi-tank fires at multiple facilities. A federal foam strategic reserve program — pre-positioned at regional locations accessible to major terminal clusters — would reduce the multi-tank escalation consequence by ensuring foam availability does not become the binding constraint on fire suppression. The National Fire Protection Association (NFPA) and the American Petroleum Institute (API) should coordinate with FEMA and DHS on the design of such a program.

7.5 Marine Terminal Navigation Closure Planning

The Scenario B consequence envelope includes a ship channel navigation closure whose duration and cross-sector consequence are not addressed in any existing emergency response plan. USCG Sector Commanders have authority to close navigable waterways to vessel traffic; the consequence of exercising that authority in a major ship channel — for all marine traffic, not only petroleum — has not been analyzed at the system level. A USCG-led consequence analysis of major ship channel navigation closures, using the Scenario B event as the design basis, should be developed to inform emergency response planning and cross-sector consequence management for the affected waterways.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the crude and product terminals and storage facility class. Issued under DBA-OIL Sector Framework v1.4. Three scenarios were developed: Scenario A (coordinated attack on Gulf Coast Colonial Pipeline origin terminal complex), Scenario B (marine terminal attack — tanker and terminal combined), Scenario C (Northeast heating oil terminal attack — seasonal timing as force multiplier). Duwal Design Basis architecture established with three terminal-specific AI governance interaction points. Regulatory gaps identified: absence of mandatory physical security standards for onshore terminals, TAS cybersecurity gap, Northeast heating oil reserve inadequacy, firefighting foam strategic reserve gap, and marine terminal navigation closure planning gap.

— END OF DOCUMENT —